RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES, MIHINTALE

B.Sc. (General) Degree

Tutorial 2 – ALGEBRA I – MAT 1207

1. For each binary operation * define on a set below, determine whether or not * gives a group structure on the set. Justify your answers.
   a) Define * on Z by a * b = ab
   b) Define * on Z by a * b = a – b
   c) Define * on $\Re^{+}$ by a * b = ab
   d) Define * on Q by a * b = ab
   e) Define * on the set of all non-zero real numbers by a * b = ab
   f) Define * on C by a * b = a + b

2. Let S be the of all real numbers except -1 {R\ -1}. Define * on S by, a * b = a + b + ab
   a) Show that * gives a binary operation on S.
   b) Show that (s, *) is a group.
   c) Find the solution of the equation, 2 * x * 3 = 7 in S.

3. Let $\Re^{*}$ be the set of real numbers except zero. Define * on $\Re^{*}$ by a * b = |a| b
   a) Show that * gives an associative binary operation on $\Re^{*}$
   b) Show that there is a left identity for * and a right inverse for each element in $\Re^{*}$.
   c) Is $\Re^{*}$ with this binary operation is a group

4. Let G: = $\{(a, b)|a \in Z, b \in Q\}$. An operation $*$ on G is defined by $(a\ b) * (c\ d) = (a + c, 2^c b + d)$
   Show that * is a binary operation and prove that (G,*) is a group.
   a) Is it abelion? What is the identity element of G?
   b) Show that H: =$\{(a, 0)|a \in Z\} \rhd G$
   c) Find another non trivial normal sub group of G.

5. On the set of H of non-zero rational numbers a binary operation * is defined by a * b = |a|b for $a, b \in H$.
   Find the element e in H, such that $e * x = x, \forall x \in H$.
   Find two elements y' and $y'' \in H$ corresponding to a given positive element $y \in H$ such that,
   $$y^* y' = y^* y'' = e$$
   Is H a group? Why?

6. G is a group with identity element e. If $(ab)^2 = (ba)^2$; $\forall a, b \in G$ and $x^2 = e \Longrightarrow x = e$. Prove that G is an abelian group.
   (Hint: Considering $(aba^{-1})^2$, Show that
   i. $ab^2 = b^2 a$
   ii. $(bab^{-1}a^{-1})^2 = (ba)^2(ab)^{-2}$)

7. Prove that a non-empty sub set H of a finite group G is a sub group iff $ab \in H$; $\forall a, b \in H$.
   A group G is generated by the two elements x and y which satisfies the relation,
   $x^3 = y^2 = (xy)^3 = 1$
   Draw the group table of G, and determine the order of G. List all the sub groups of G.

8. Let Z be a set of integers. Define operator "*" on Z such that $a * b = a - 3 + b$; $a, b \in Z$
   i. Show that (Z,*) is a group.
   ii. What is the identity element of (Z,*)?
   iii. Is (Z,*) cyclic?

iv. If so, find a generator of (Z,*)

9.

a) A function f is defined by $f(x) = (10 - k^2)x^2 - (k - 4)x + k$, where k is a given integer. For what values of k is f injective?

b) Let n be a given integer, and let $f: Z \to Z$ be the function defined by $f(x) = x + n - xn$. Find the values of n for which f is

i. injective

ii. surjective, also find f^{-1} when f is bijective.

10. Determine whether each of the following * is a binary operation on the respective set. Also, if * is a binary operation determine whether it is associative and/or communicative.

i. On R defined by $a * b = a - b$

ii. On Z^+ defined by $a * b = a^b$

iii. On $R - \{1\}$ defined by $a * b = a + b - ab$

iv. On Z^+ defined by $a * b = 2^{ab}$

v. On Z^+ defined by $a * b =$the smallest integer less than both a and b.

vi. On Q defined by $a * b = ab/2$

11. Let S be a non-empty set and * be an associative binary operation on S. If S has an element u such that $u * x = x$ for all x in S, and for each $a \in S$, there is an element $a' \in S$ such that $a' * a = u$, prove that

i. If $x \in S$ and $x * x = x$, then $x = u$.

ii. If x and y are in S and $y * x = u$, then $x * y = u$.

iii. $x * u = x$ for all $x \in S$.

iv. u is unique.

v. If $a * a' = a'' * a = u$ for $a, a', a'' \in S$, then $a' = a''$

END